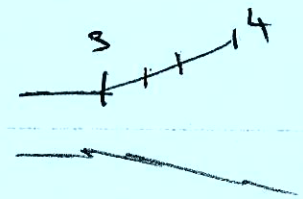
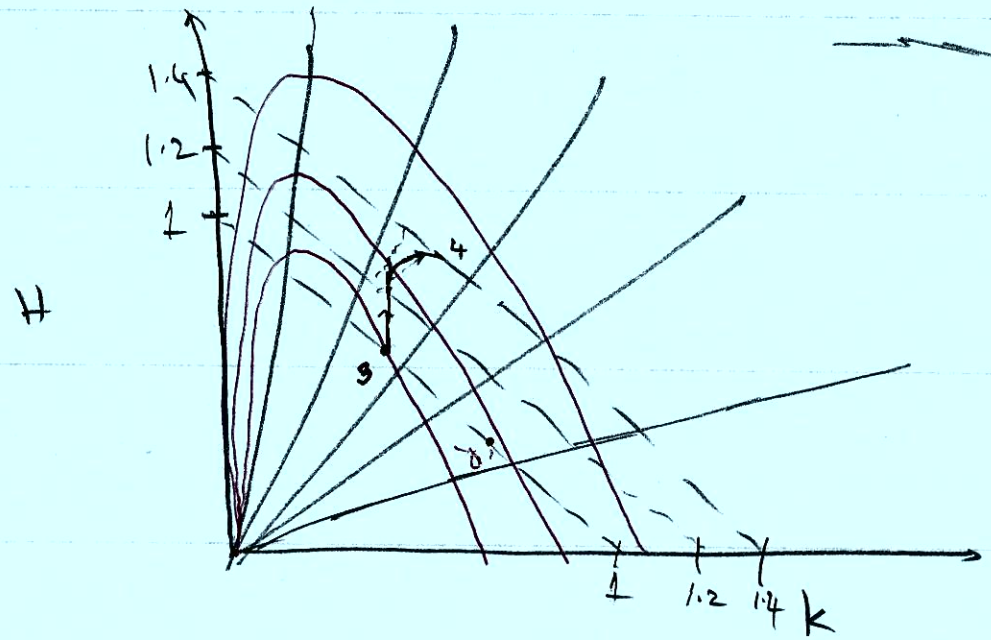
□ H-k diag. of shock-free (no isol. shocks) soranyet op.





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Concept of dual mode combustion system.

- "Pure" ramjet engine needs two physical throats, one to locate a normal shock @ inlet exit ~~plate~~ plane to deliver $M_3 < 1$ to the burner & other downstream of burner exit to deliver supersonic flow.
- Pure scramjets needs/has no physical throat.
- We would like to operate in either/both ramjet or scramjet mode only using a "no-throat" geometry – How?
- Transition from ram/scram is accomplished in dual-mode engine by means of isolator.
  - In ramjet mode, $M_3 < 1 \Rightarrow$ normal shock train in the isolator $\Rightarrow$ large p_3 @ burner entry.
  - p_3 determined by the need to have a <u>thermal choking</u> d/s of the burner exit.
- Dual mode (Ram mode) $\rightarrow$ go for thermal choking ~~to 2nd throat~~ instead of 2nd throat.
  $\rightarrow$ use N.S. train in the isolator to bring $M_3 < 1$ & meet p_3 requirement.
- In scram mode, we don't need nad an isolator as there is no need to get $M = 1$ anywhere in the engine.
  - But the isolators help containing the p_3 excursions during scram operation.

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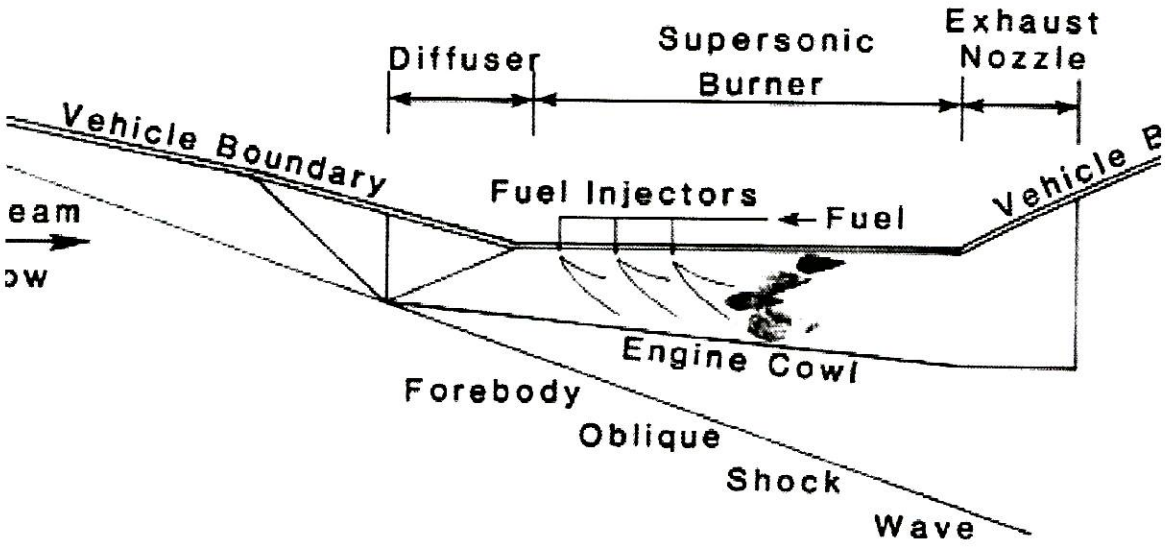
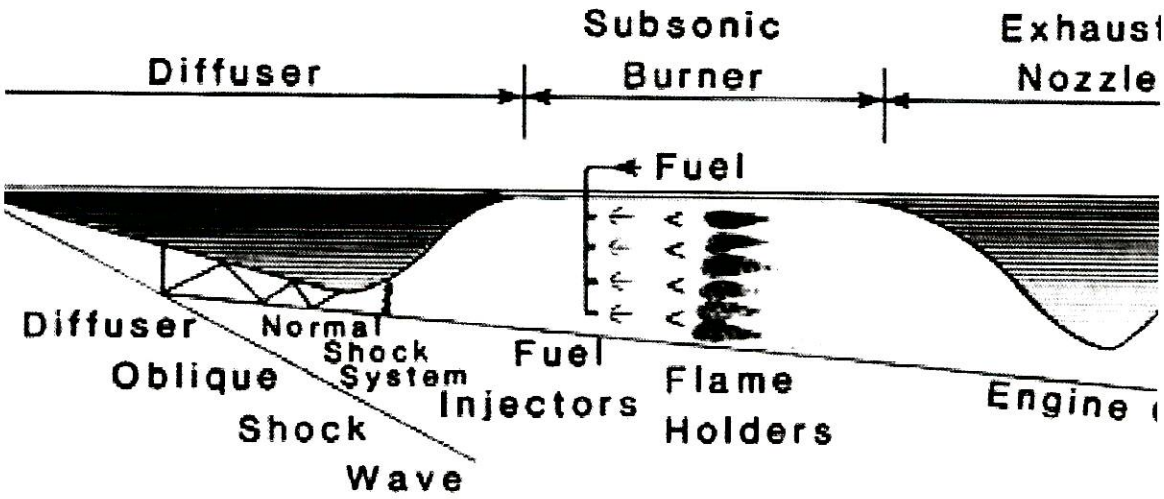
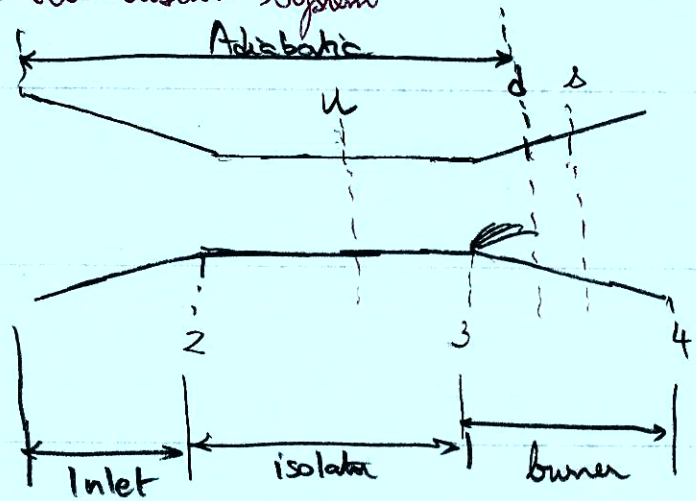


Schematic diagram of a two-dimensional or pla
mjet engine.



ıematic diagram of a two-dimensional or p
engine.

Aerothermodynamics of the Combustion System



3–4 : burner entry thru exit

2 : entry to the isolator

3 : isolator exit / burner entry

u–d : u/s & d/s limits of positive (adverse) pr grad

d : start of heat add^n. (due to comb)

d–s : region of pr plateau

s–4 : pr. decreases ⇒ fav. pr. grad.

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3) Chemical Kinetics.

- Reactions take a finite amount of time & occur @ a certain rate.

Non-equil effects dominate high-speed prop. because residence time is very short

$$A + B \xrightarrow{k} C.$$

k = rxn rate

$$k_j = A_j T^{\beta_j} \exp\left(\frac{-E_a}{R_u T}\right)$$

A_j : f(concentration)

For simplest fuel (H_2), a good mech uses 13 species & 33 steps.

For " HC (CH_4), 30 " " 100 steps.

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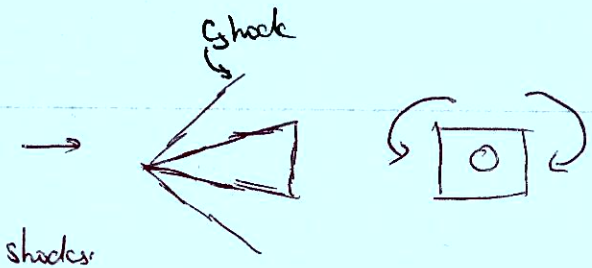
Axial vortex mixing

- Want to use vortical structures to lift the fuel up to freestream as quickly as possible without causing shocks.



- Fuel entering combustor will have some momentum.
- We want to recover fuel momentum as much as possible ⇒ inject the fuel parallel to flow path while minimizing shock losses.

Chemical Kinetics.

Levels of complexity (kinetics)

1) Stoichiometric comb. with saturated products & inerts

$$H_2 + \frac{1}{2}\left(O_2 + 3.76\,N_2\right) \longrightarrow H_2O + 1.88\,N_2$$

2) Equilibrium with dissociation and nitrogen chemistry

$$H_2 + \frac{1}{2}\left(O_2 + 3.76\,N_2\right) \longrightarrow a\,H_2O + b\,O + c\,OH + d\,O_2 + e\,H + f\,H_2O_2 + g\,HO_2 + h\,NO + i\,NO_2 + j\,N_2O + l\,N$$

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